

Scrambling, not athermality, protects quantum information under generic measurement: chaotic states outperform quantum many-body scar codes

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Monitored quantum many-body systems protect quantum information from projective measurements exactly when their unitary dynamics scramble it, placing the measurement-induced entanglement transition on the same footing as quantum error correction. Because the eigenstate thermalization hypothesis endows chaotic eigenstates with approximate Knill–Laflamme structure, it is natural to ask whether the *non-thermal* eigenstates known as quantum many-body scars—atypically weakly entangled and dynamically coherent—form especially good codes. Whether scars protect quantum information better or worse than the surrounding thermal states under random measurement has not been settled quantitatively. Here we show that scrambling, not athermality, is what protects an encoded qubit: under monitored dynamics, the reference-qubit coherent information of a logical qubit encoded in the PXP scar subspace lies below that of an energy-matched thermal encoding for every measurement we test. The scar disadvantage is mild for local measurement, is largest for collective measurements aligned with the emergent $\text{su}(2)$ order parameter (which couples the code states), and—decisively—survives even a fine-tuned Casimir measurement for which the scar code is a nominal decoherence-free subspace, because approximate scars carry irreducible Casimir variance. These results confirm the eigenstate-thermalization-to-error-correction correspondence directly in the monitored-circuit setting and refute the hypothesis that scarring aids information protection against generic measurement. A second, exactly-solvable spin-1 model isolates the Casimir variance as the control parameter: only zero-variance *exact* scars are protected by their Casimir, and even then a generic local measurement grants them no advantage. These results establish scrambling as the design resource for measurement-robust encoding and caution that weak-ergodicity-breaking subspaces are liabilities rather than assets for measurement-based quantum memories.

I. INTRODUCTION

The competition between scrambling unitary dynamics and local measurements drives a sharp transition in how much quantum information a many-body system retains [1–7]. This measurement-induced phase transition (MIPT) is now understood as a coding transition: below a critical measurement rate the dynamics act as an emergent quantum error-correcting code that hides logical information non-locally, protecting it from measurements that act as erasure errors [8–10]. The order parameter is information-theoretic—the coherent information, or equivalently the entanglement of a reference qubit maximally entangled with the encoded information [9, 11, 12].

The coding power of chaotic dynamics is inherited by chaotic *eigenstates*. The eigenstate thermalization hypothesis (ETH) forces local operators to have smooth diagonal and exponentially small off-diagonal matrix elements [13–16], which is exactly the approximate Knill–Laflamme condition [17]; thermal eigenstates therefore form approximate codes whose error scales with the system’s chaoticity [18–20]. Quantum many-body scars are the conspicuous exception to ETH: a measure-zero tower of atypically weakly entangled, dynamically reviving eigenstates embedded in an otherwise thermal spec-

trum [21–25]. Scars can be embedded in decoherence-free subspaces [26–28], and projective measurement can even enhance their revivals [29], suggesting they might be privileged carriers of quantum information under monitoring.

However, whether a logical qubit stored in the scar subspace is protected better or worse than one stored in the surrounding thermal bulk—at matched energy, in the same monitored dynamics, quantified as a channel capacity—has not been determined. In this study, we compute the reference-qubit coherent information $\mathcal{C}_R$ of scar and thermal codes of the monitored PXP chain across local, collective, and Casimir measurement models. We find that thermal codes protect information better than scar codes in every case; that an apparent scar advantage seen against a single thermal reference is an artifact of an atypical baseline and vanishes under ensemble averaging; and that the scar’s emergent- $\text{su}(2)$ structure is a liability, not a shield, because its generator couples the code states and the scars’ Casimir variance defeats even a fine-tuned decoherence-free encoding.

The result establishes the ETH-to-error-correction correspondence as an operational statement about monitored circuits and removes weak-ergodicity-breaking subspaces from the menu of candidate measurement-robust quantum memories.

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II. MODEL AND DIAGNOSTICS

We study the PXP chain of L sites in the Rydberg-blockaded (constrained) Hilbert space of dimension $F(L+2)$ (Fibonacci), with

$$H = \sum_i P_{i-1} X_i P_{i+1}, \quad P = |0\rangle\langle 0|, \quad (1)$$

whose $\mathbb{Z}_2$ Néel state exhibits the canonical scar revivals [21, 22, 30, 31] (period $t \simeq 4.7$, fidelity $\simeq 0.74$ at $L = 12$). We identify the $L+1$ scar-tower eigenstates by their anomalous overlap with $|\mathbb{Z}_2\rangle$.

A logical qubit is encoded in a two-dimensional subspace $\{|0_L\rangle, |1_L\rangle\}$ and maximally entangled with a reference R , $|\Psi_0\rangle = (|0\rangle_R |0_L\rangle + |1\rangle_R |1_L\rangle)/\sqrt{2}$. The system evolves by alternating steps of PXP evolution $e^{-iH dt}$ and random projective measurement; per quantum trajectory the joint state stays pure and the reference entanglement entropy $S_R = -\text{Tr} \rho_R \log_2 \rho_R \in [0, 1]$ measures the surviving logical information. Its trajectory average

$$\mathcal{C}_R(p) = \langle S_R \rangle_{\text{traj}} \quad (2)$$

is the single-qubit coherent information of the monitored channel: $\mathcal{C}_R \rightarrow 1$ recoverable, $\mathcal{C}_R \rightarrow 0$ lost. We take $dt = 0.6$ and measurement probability p per site (local) or per operator (collective) per step. To remove the energy-mismatch confound, scar and thermal codes are built from eigenstates at the *same* two energies; **the thermal baseline is averaged over an ensemble of ~ 10 – 20 energy-matched thermal pairs**, without which a single atypical pair inverts the conclusion (Sec. III).

We study $L = 10$ – 18 (constrained dimension up to $F(20) = 6765$) with 40 evolution–measurement steps per trajectory and average $\mathcal{C}_R$ over 120–400 Born-sampled quantum trajectories. Reported statistical error bars are the standard error of the mean over trajectories; the thermal band is the standard deviation of $\mathcal{C}_R$ over the ensemble of energy-matched thermal pairs. All runs use fixed random seeds and are reproducible from the accompanying code.

Statically we probe the Knill–Laflamme violation of a code under an error set $\{E_a\}$ through the logical leak $\langle 0_L | E_a | 1_L \rangle$ and the distinguishability $\langle 0_L | E_a | 0_L \rangle - \langle 1_L | E_a | 1_L \rangle$.

a. A model-independent criterion. A projective measurement of a Hermitian operator O leaves the encoded qubit literally untouched only if the code is an eigenspace of O with a single shared eigenvalue; more generally, the qubit survives a single shot if and only if the Knill–Laflamme conditions hold for the projector set $\{\Pi_g\}$ of O ’s eigenspaces [17]. A measurement *resolves*—and hence degrades—the code through two channels: (i) *eigenvalue distinguishability*, when the two logical states’ outcome distributions $P_a(g) = \langle a_L | \Pi_g | a_L \rangle$ differ; and (ii) *non-invariance*, when a sector-restricted overlap $\langle 0_L | \Pi_g | 1_L \rangle$ is nonzero, coherently mixing the logical states. Equal means and a vanishing global leak

$\langle 0_L | O | 1_L \rangle$ are therefore insufficient: a code spread over many O -sectors—a large O -variance—is collapsed by the shot, and survival then requires the sector-wise conditions in *every* occupied sector, which is why a true decoherence-free subspace demands logical states that are O -eigenstates sharing one eigenvalue. This criterion is independent of the microscopic model and organizes the results below: thermal states, spread broadly and near-identically in a local basis, are close to optimal codes, while scars are penalized whenever the measured operator resolves their atypical structure. Both channels combine into a single exact, simulation-free scalar—the reference entropy $s_1(O)$ surviving one O -measurement, which equals unity precisely when the sector-wise conditions hold—and s_1 predicts the monitored $\mathcal{C}_R$ across resolving measurements (Supplement).

III. RESULTS

a. Thermal codes win; the apparent scar advantage is a baseline artifact. Figure 1 compares $\mathcal{C}_R(p)$ for the scar code against the thermal ensemble. For both local- Z and collective (staggered- Z) monitoring the scar curve lies *below* the thermal-ensemble band at all p . A single nearest-in-energy thermal pair sits at the 0th percentile of the ensemble ($\mathcal{C}_R = 0.53$ versus ensemble mean 0.87 at $L=14$, $p=0.10$, collective), which is why an unaveraged comparison spuriously favors the scar.

b. The disadvantage persists to the largest accessible size. Figure 2 shows $\Delta\mathcal{C}_R \equiv \mathcal{C}_R^{\text{scar}} - \langle \mathcal{C}_R^{\text{therm}} \rangle$ at $p = 0.10$. It is negative for all $L = 10$ – 18 and both channels: -0.03 to -0.06 (local) and -0.25 to -0.32 (collective), the band excluding zero. At the largest size $L = 18$ (constrained dimension 6765) it is -0.05 (local) and -0.30 (collective), essentially size-independent: the refutation is not a finite-size artifact.

We emphasize that this single-qubit comparison is the most favorable one for the scar: the scar tower is only $(L+1)$ -dimensional, so it can host at most $\log_2(L+1)$ logical qubits—a sub-extensive capacity, against the exponentially large thermal bulk. A scar subspace therefore cannot support an extensive code even before dynamics; the per-qubit comparison already settled here is its best case, and it loses.

To confirm this directly, we encode the *maximal* scar code—all $k = \lfloor \log_2(L+1) \rfloor$ logical qubits, i.e. 2^k tower states maximally entangled with a k -qubit reference—and compare its protected-information density S_R/k against a same-dimension thermal code under local monitoring (Fig. 3). The scar loses at every rate: at $L = 14$ ($k = 3$), $\Delta(S_R/k) = -0.17, -0.14, -0.06$ at $p = 0.02, 0.04, 0.08$, with the thermal ensemble tightly determined ($\sigma \lesssim 0.01$), and the gap persists across $L = 10, 14, 18$. The gap is even larger under collective staggered- Z monitoring ($\Delta(S_R/k) = -0.32$ to -0.43 at $L = 14$), while under the fine-tuned Casimir J^2 the two codes are comparable ($|\Delta| \lesssim 0.03$; Supplement)—the

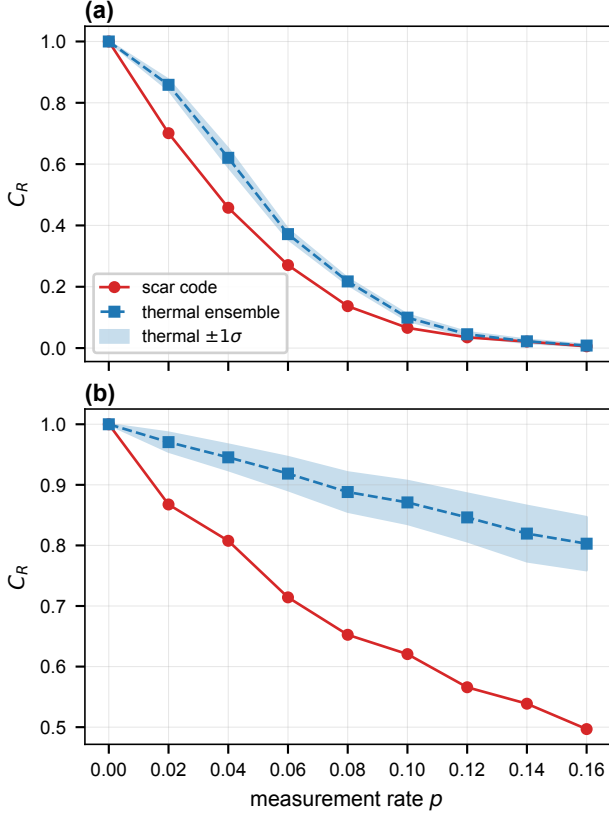


FIG. 1. **Thermal codes protect information better than scar codes.** Reference-qubit coherent information C_R versus measurement rate p for the scar code (red) and the thermal ensemble (blue; band is $\pm 1\sigma$ over the ensemble) under (a) local-Z and (b) collective staggered-Z monitoring at $L = 14$. The scar lies below the thermal band throughout.

same exception seen in the single-qubit and spin-1 analyses. Even the largest code the scar subspace admits is thus a worse monitored memory than a generic thermal code of equal size under any generic measurement.

c. *Mechanism: both channels resolve the scar code.* The PXP scar realizes both channels of the criterion while thermal codes realize neither. Channel (ii) dominates under the collective staggered operator $M = \sum_i (-1)^i Z_i$ —the staggered magnetization, order parameter of the emergent $\text{su}(2)$ scar dynamics [32]—which carries a large logical leak $|\langle 0_L | M | 1_L \rangle| = 1.73$ between the scar rungs (versus ≈ 0 for thermal codes) and coherently mixes them. This leak is a one-sided suppressor of C_R [Fig. 4(b)]: every high-leak code is poorly protected, whereas low leak permits any value. Channel (i)—eigenvalue distinguishability through O -variance—is isolated by the Casimir measurement below. Accompanying both is a weak monotone trend [Fig. 4(a)]: protection rises with the code’s spread in the measurement basis, as ETH predicts, so that unstructured thermal codes are close to optimal.

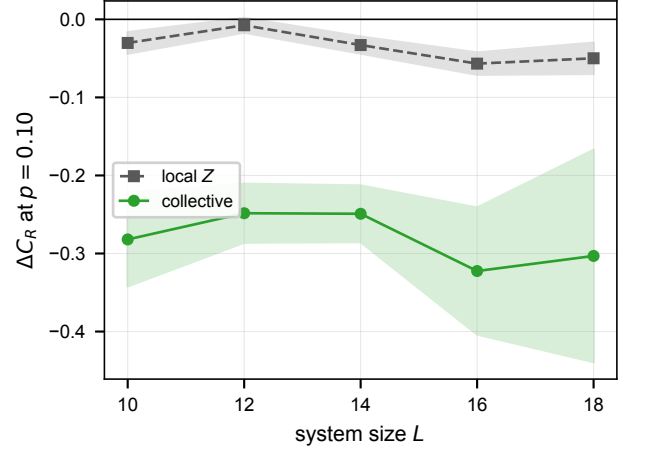


FIG. 2. **The scar disadvantage $\Delta C_R < 0$ at every size.** Scar-minus-thermal coherent information at $p = 0.10$ versus L for local (grey) and collective (green) monitoring; shading is the thermal-ensemble spread.

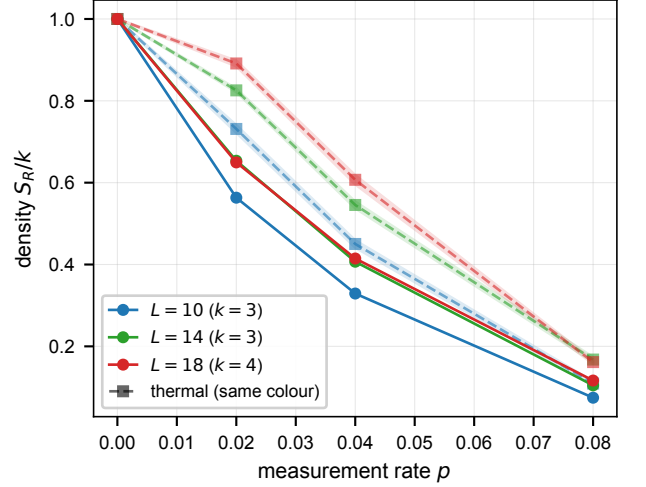


FIG. 3. **The maximal scar code loses even as an extensive memory.** Protected-information density S_R/k for the maximal $k = \lfloor \log_2(L+1) \rfloor$ -qubit scar code (solid) and a same-dimension thermal code (dashed, $\pm 1\sigma$ band) versus p under local monitoring, for $L = 10, 14, 18$. Scar lies below thermal throughout.

The full (p, ℓ) dependence, with ℓ the measurement support size (block-sum of ℓ sites; $\ell=1$ local, $\ell=L$ global), is negative throughout: the scar never wins (Supplemental Fig.).

d. *The decoherence-free-subspace rescue fails.* A true DFS [26, 27] for the scar code would restore an advantage. The emergent Casimir $J^2 = \frac{1}{2}(H^+ H^- + H^- H^+) + J_z^2$, built from the exact split $H = H^+ + H^-$ [32], gives the chiral-symmetric scar pair $(|+E\rangle, |-E\rangle)$ equal expectation and vanishing leak

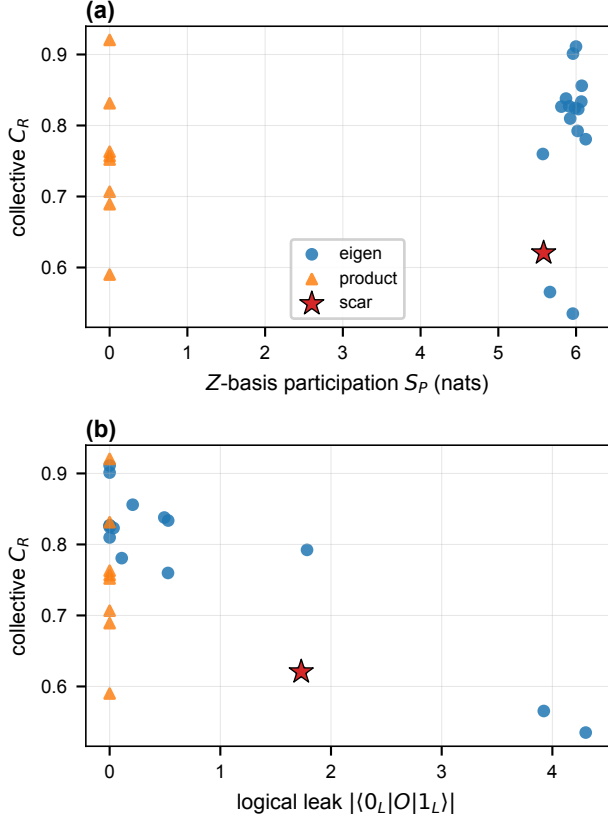


FIG. 4. **What controls collective protection.** Collective C_R versus (a) Z-basis participation and (b) logical leak $|\langle 0_L | O | 1_L \rangle|$ over an eigenstate sweep (blue), product states (orange), and the canonical scar (star). Higher participation raises C_R ; large leak suppresses it. The scar is a low- C_R outlier on both axes.

($\Delta \langle J^2 \rangle = 0.000$, leak = 0.052)—a nominal DFS. Yet under J^2 monitoring the scar code still loses, $\Delta C_R = -0.18, -0.13, -0.09$ at $p = 0.04, 0.08, 0.12$. The reason is that equal expectation and zero leak are necessary but not sufficient: a genuine DFS requires the code states to be *eigenstates* of the measured operator. Approximate PXP scars carry irreducible Casimir variance ($\text{Var } J^2 = 32.5$ versus 18.1 for the thermal state), so J^2 measurement collapses them more strongly than it does thermal states. No Casimir measurement can be a true DFS for approximate scars.

e. A second model isolates the variance as the culprit. To test generality and to isolate this mechanism we repeat the analysis in the spin-1 Schechter–Iadecola magnet [33], whose scar tower $|S_n\rangle = (Q^+)^n |\Omega\rangle$, $Q^+ = \sum_j (-1)^j (S_j^+)^2$, is an *exact* $\text{su}(2)$ multiplet. Two facts then hold to machine precision: the scars are exact eigenstates, and the Casimir J^2 is exactly constant on the tower with *zero* variance on every scar—against $\text{Var } J^2 \simeq 32$ for PXP [Fig. 5(b)]. This flips the Casimir rescue [Fig. 5(a)]: the exact- $\text{su}(2)$ scar code is now a gen-

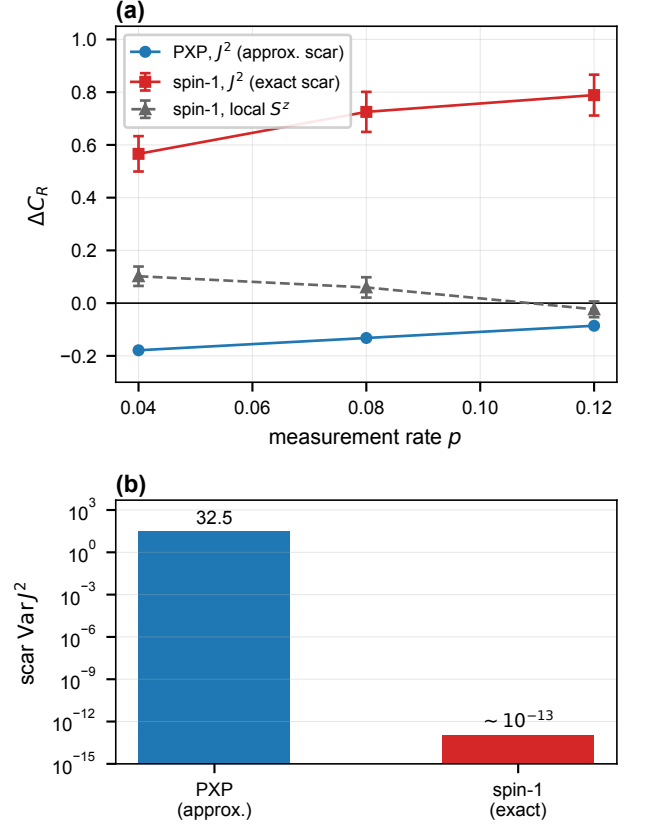


FIG. 5. **Exact- $\text{su}(2)$ scars evade the Casimir-variance obstruction.** Second model (spin-1 Schechter–Iadecola, $L = 6$, 20-pair thermal ensemble). (a) Scar–thermal coherent information ΔC_R versus p (error bars are the ensemble SEM). Under J^2 monitoring PXP scars lose (blue) while exact spin-1 scars win perfectly (red); under generic local S^z the exact-scar and thermal codes are comparable ($|\Delta C_R| \lesssim 0.1$, grey). (b) Scar Casimir variance, $\simeq 32$ (PXP) versus $\sim 10^{-13}$ (exact) on a log scale—the control parameter of the rescue.

uine decoherence-free subspace—its two logical states are identical J^2 eigenstates—and is *perfectly* protected under J^2 monitoring, $C_R^{\text{scar}} = 1$ at every p , giving $\Delta C_R = +0.6$ to $+0.8$ ($L = 6, 8$). The rescue that fails for approximate PXP scars succeeds exactly when the Casimir variance vanishes, confirming that this variance—not scarring itself—controls it. This is parameter-independent: for $(h, D) = (0.5, 0.3)$ and $(1.5, 0.0)$ the Casimir variance again vanishes identically and $C_R^{\text{scar}} = 1$ under J^2 . The advantage is, however, confined to this fine-tuned Casimir probe: under generic local S^z monitoring the exact-scar and thermal codes are statistically comparable ($|\Delta C_R| \lesssim 0.1$), so scar structure still confers no advantage against a generic local measurement.

f. Calibration to prior monitored-scar work. Reproducing the monitored-Néel entanglement transition and mapping our per-step rate to continuous time via $\gamma = p/dt$ yields a critical rate $\gamma_c \simeq 0.083$ ($p_c \simeq 0.05$), the same order as the $\gamma_c \simeq 0.013 \pm 0.002$ reported for continuous-

time projective σ^z monitoring of PXP scars [29]; the factor ~ 6 is consistent with the per-step versus continuous-time rate convention.

g. Experimental accessibility. The negative result is testable on the platform where PXP scars were first observed [21]. Programmable Rydberg-atom arrays natively realize the blockaded dynamics, local- Z monitoring is site-resolved projective readout interleaved at a tunable rate, and the sizes studied here ($L \leq 18$) are modest by current standards. Encoding in the scar subspace is accessible through the Néel revival manifold, and the retained coherence can be estimated by interferometric or randomized-measurement protocols. While preparing individual thermal eigenstates is impractical, thermal-like codes can be prepared operationally by pre-scrambling product states under the chaotic dynamics itself, and the scar code's excess fragility appears directly as a faster purification of the Néel-manifold encoding.

IV. CONCLUSION

We quantified, as a monitored-channel coherent information, how well a logical qubit stored in the PXP quantum-many-body-scar subspace is protected against random measurement, benchmarked against an ensemble-averaged thermal code at matched energy—to our knowledge the first such head-to-head coding comparison of scar versus thermal subspaces under monitoring. Thermal codes protect information better than PXP scar codes at every size and under every measurement model, by $\Delta\mathcal{C}_R \simeq -0.05$ (local) to -0.3 (collective) at $p = 0.10$, up to $L = 18$; the scar's emergent- $su(2)$ order parameter supplies a logical leak of 1.73 that makes it especially fragile, and even a fine-tuned Casimir DFS fails because approximate scars retain Casimir variance ~ 30 . This realizes the eigenstate-thermalization-to-approximate-error-correction correspondence [18–20]

as an operational statement about monitored circuits: chaos, not athermality, is the coding resource. Any program that would exploit weak ergodicity breaking for measurement-robust quantum memory should expect to lose to a generic chaotic subspace; the relevant resource is the code's spread in the measurement basis, and structured non-thermal towers whose generators connect the code states are actively harmful. A second, exactly-solvable spin-1 model [33] whose tower closes the algebra exactly confirms the obstruction is variance-controlled—there the Casimir measurement protects the scar perfectly, while under generic local measurement scars retain no advantage. Whether this fragility survives generic non-projective noise channels and finite-temperature encodings [25, 34] is the natural next question.

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AUTHOR CONTRIBUTIONS

H.W. and T.T. conceived the study. H.W. developed the methodology, implemented the software, performed all computations and analysis, produced the figures, and wrote the manuscript.

DATA AND CODE AVAILABILITY

The simulation package, drivers, and the data underlying all figures are openly available at [doi:10.5281/zenodo.21336840](https://doi.org/10.5281/zenodo.21336840) (source: <https://github.com/hwakaure/scarcodes>), enabling one-command reproduction of every result reported here.

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